Easy Visa

Ensemble techniques and Model Tuning

13th Sept 2025

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- Model Performance Summary
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Executive Summary

Actionable insights and Recommendations

- Factors that influence visa certification:
  - Higher education-
    - Bachelor's – 26% increase
    - Master's – 17% increase
    - Doctorate – 8% increase
  - High prevailing wage - more likely to get certified
  - Previous work experience -20% increase in getting certified. Recommendations
- Visa applicants with only Highschool education have a higher chance of denial.
- Visa applicants with prevailing wage, work experience and higher education have a much better chance of acceptance/certifications.

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Business Problem Overview and Solution Approach

Problem Overview

In 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications. Processed applications increased by nine percent YOY.

The process of reviewing every case is becoming a tedious task.

A machine learning based solution can help in shortlisting the candidates having higher chances of VISA approval. OFLC has hired Easy Visa for data-driven solutions.

Solution approach

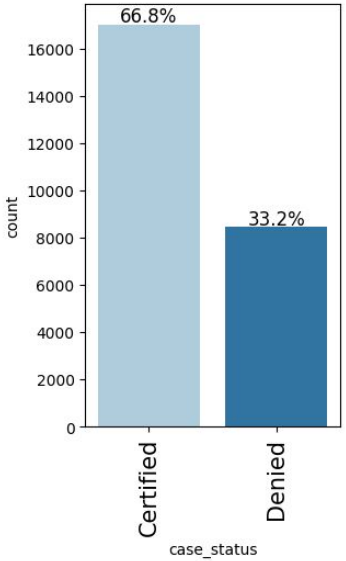
Easy Visa must analyze the data provided. This can -

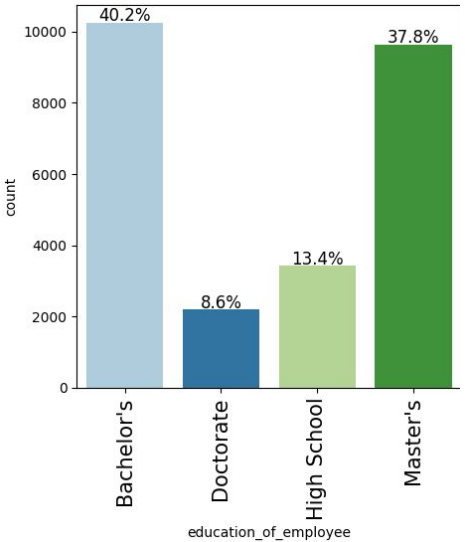
1. Facilitate the process of visa approvals.

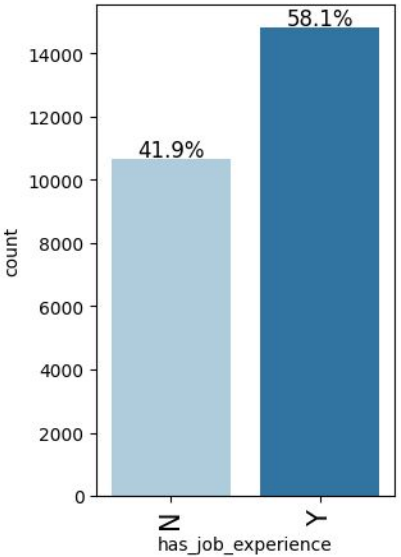
2. Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

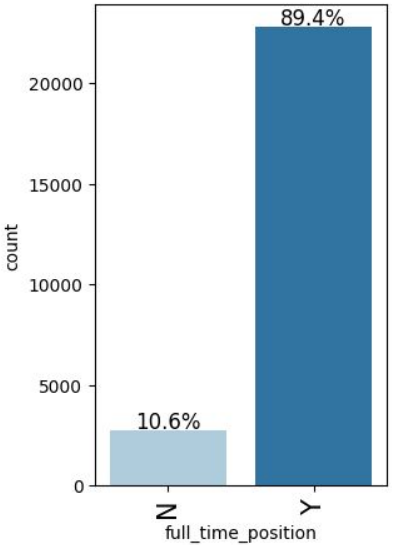
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EDA Results - Univariate analysis





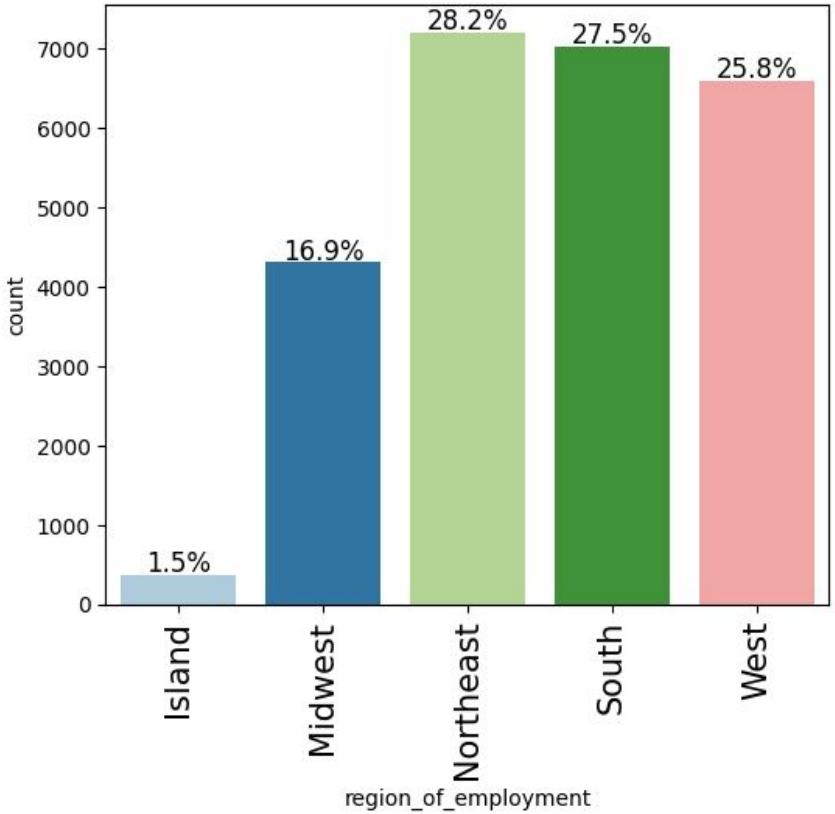




- ⅔ of all visa applicants are certified.
- Most visa applicants have a bachelor's or master's degree.
- 58% of all visa applicants have previous job experience.
- 89% of all visa applicants have a full time job.

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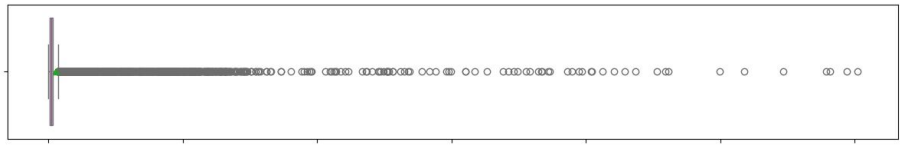
EDA Results - Univariate analysis

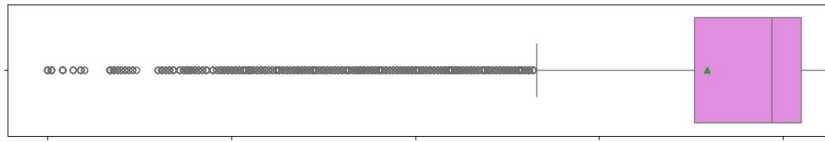


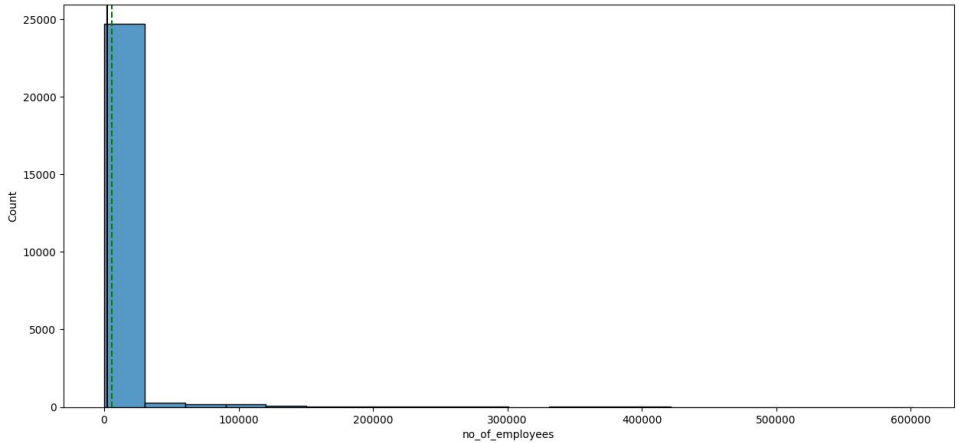
Most visa applicants are from Northeast, South and West regions. They comprise of 80% of all visa applicants.

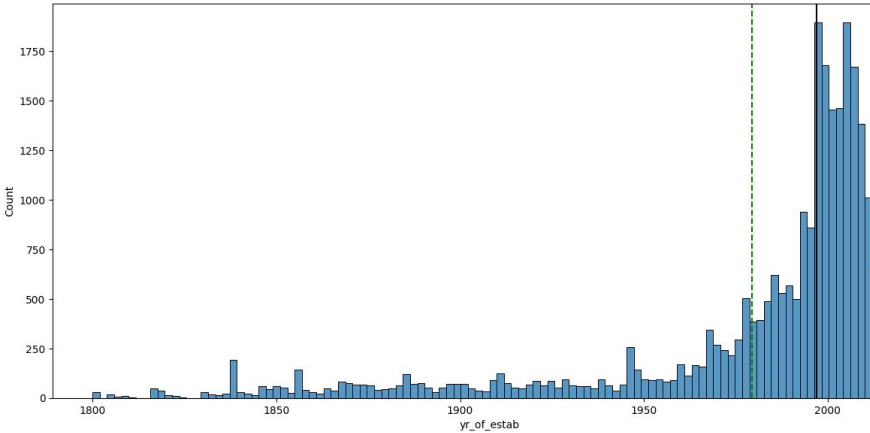
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EDA Results - Univariate analysis





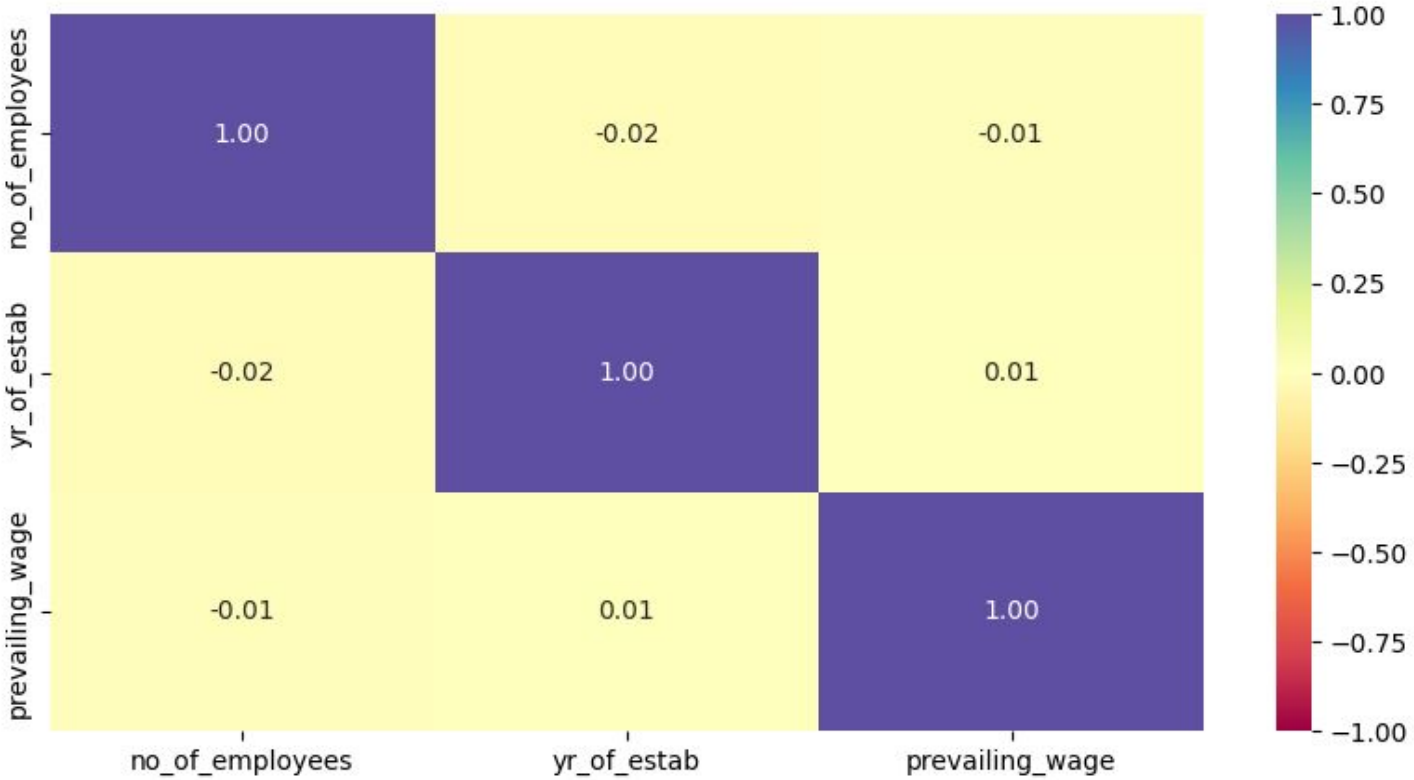




Year of establishment is after 1975 for most visa applicants.

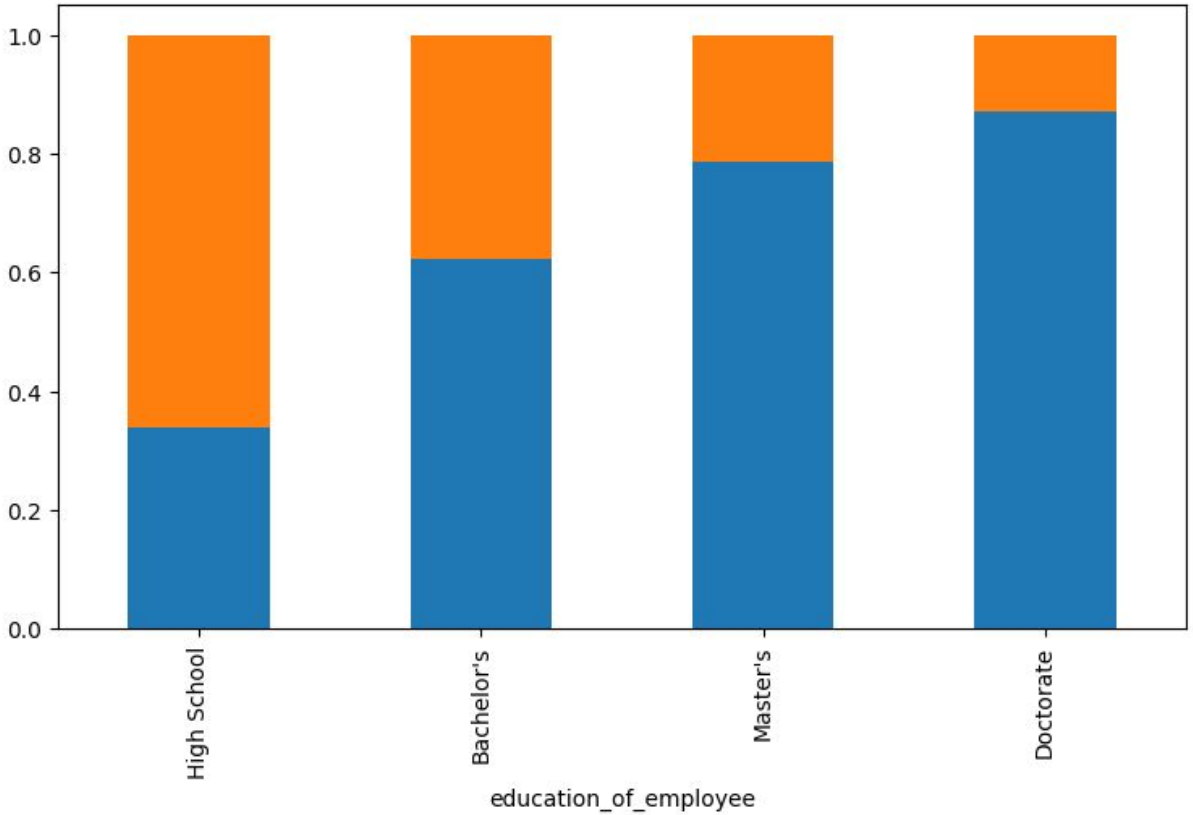
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EDA Results - Bivariate analysis



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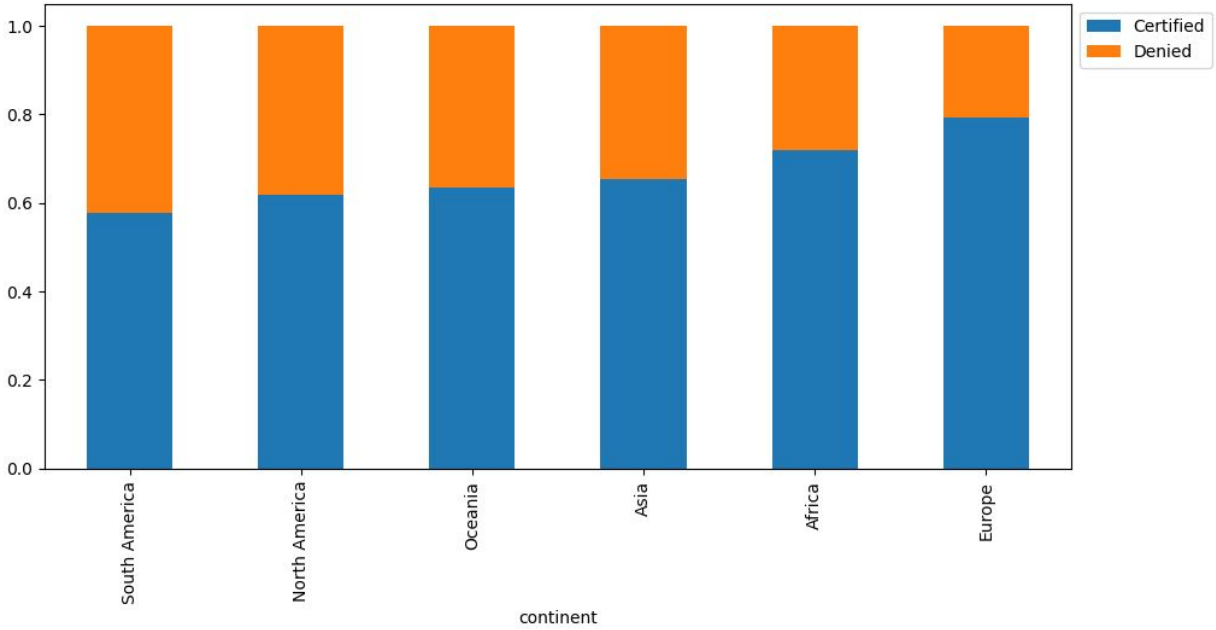
EDA Results - Bivariate analysis - Education of employee



VISA applicants with a doctorate have the highest certification rate - 87%. Master's – 79%
Bachelor's – 62%
High School – 36%

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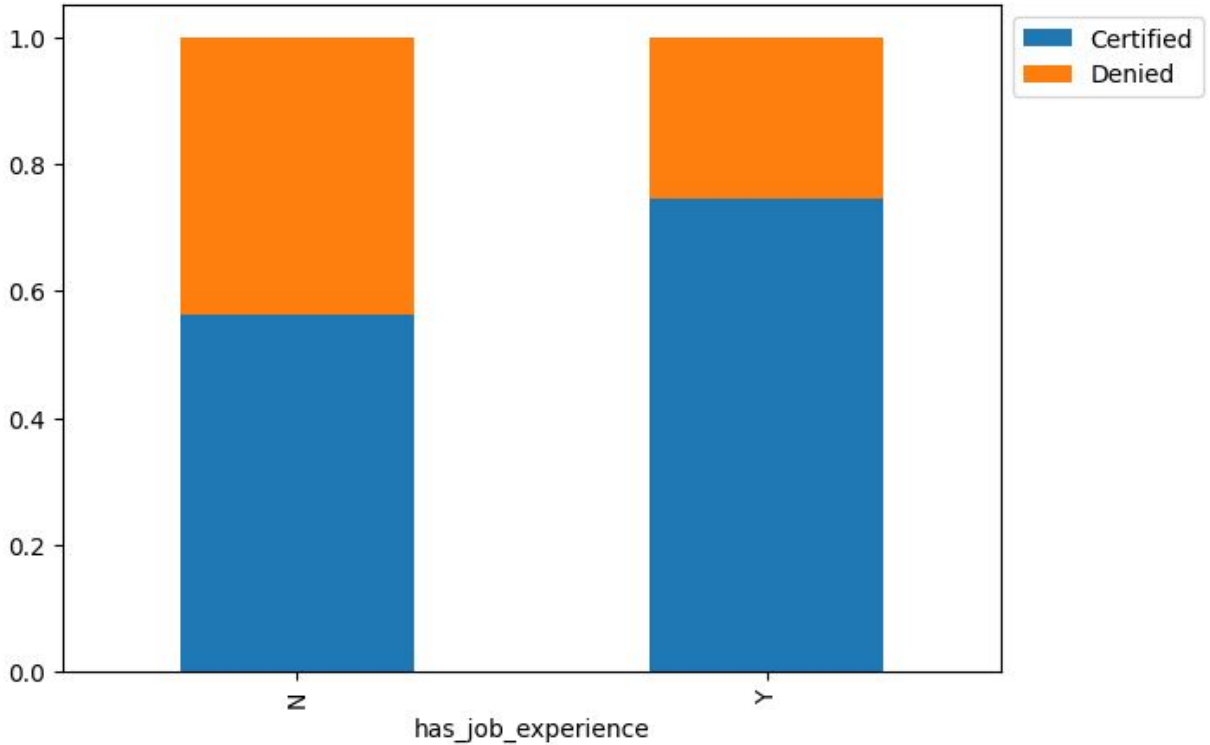
EDA Results - Bivariate analysis - Case Status Vs Continent



The Midwest region has the highest Visa application certification rate -77%
South – 70%
Northeast – 62%
West – 61%
Island – 60%

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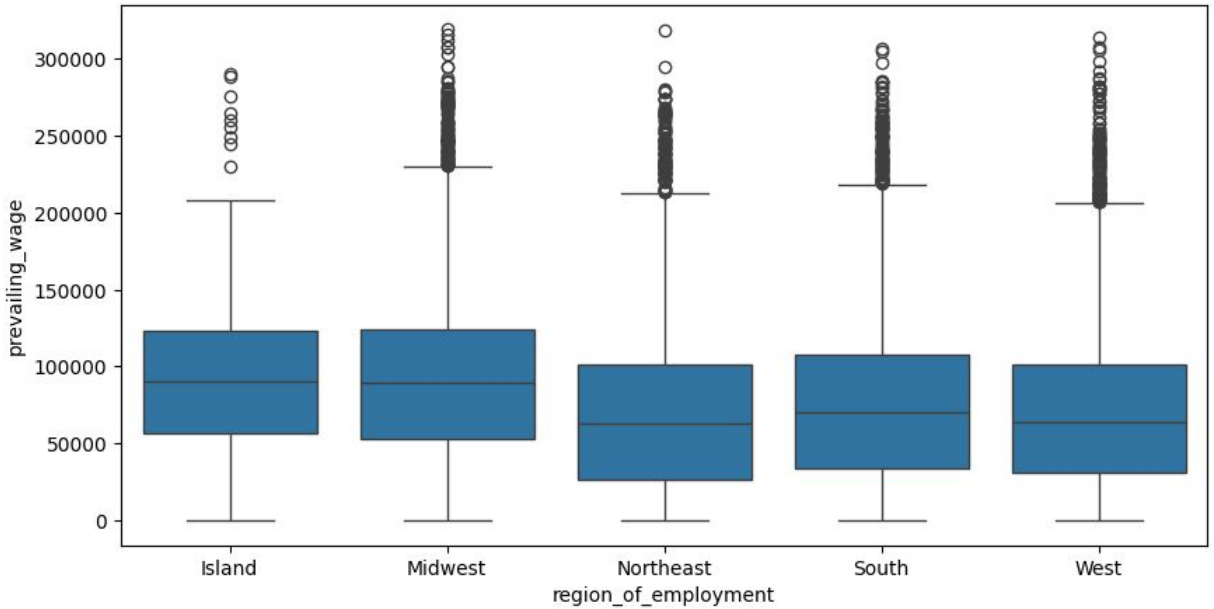
EDA Results - Bivariate analysis - Case Status Vs Job Experience



1. VISA applicants who do have previous work experience have a higher certification rate of 77%.
2. VISA applicants with no previous work experience have a certification rate of 58%
3. VISA candidates with previous work experience show a 20% increase in getting certified.

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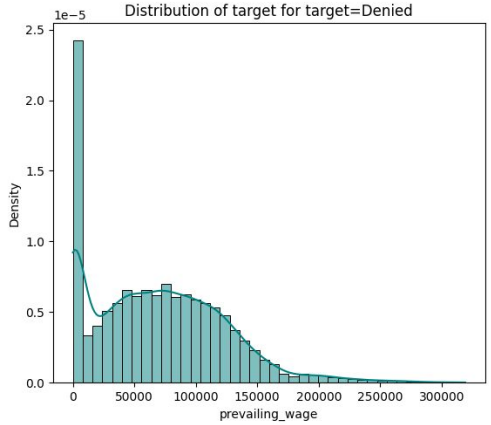
EDA Results - Bivariate analysis - Prevailing wage Vs Region

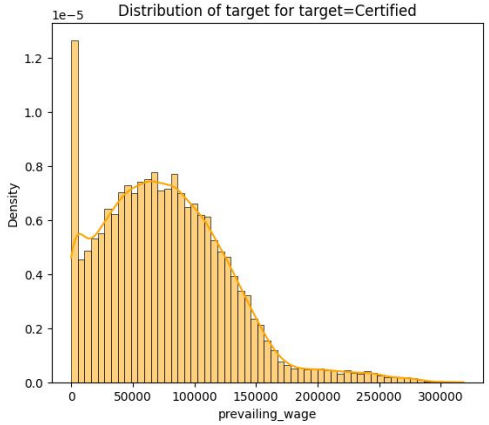


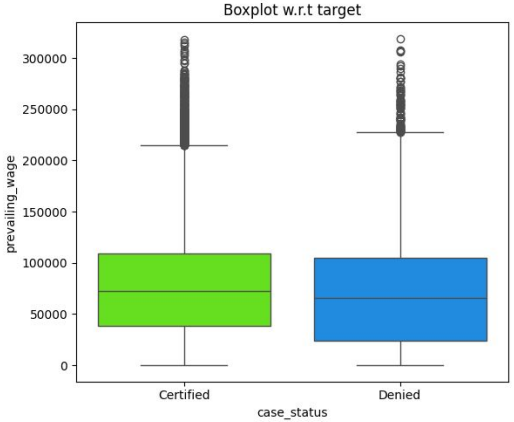
Island and Midwest have higher mean prevailing wage than other regions of employment.

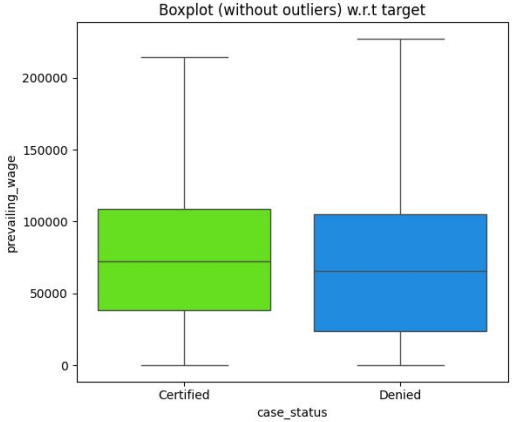
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EDA Results - Bivariate analysis - Visa Status Vs prevailing wage





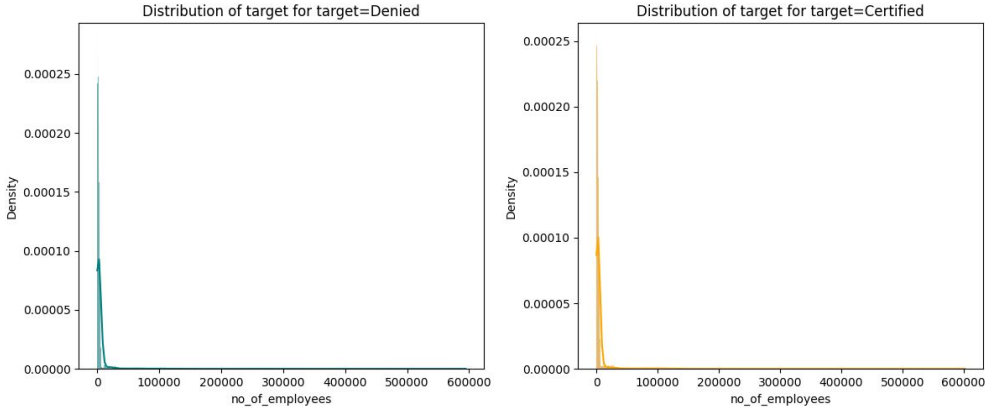


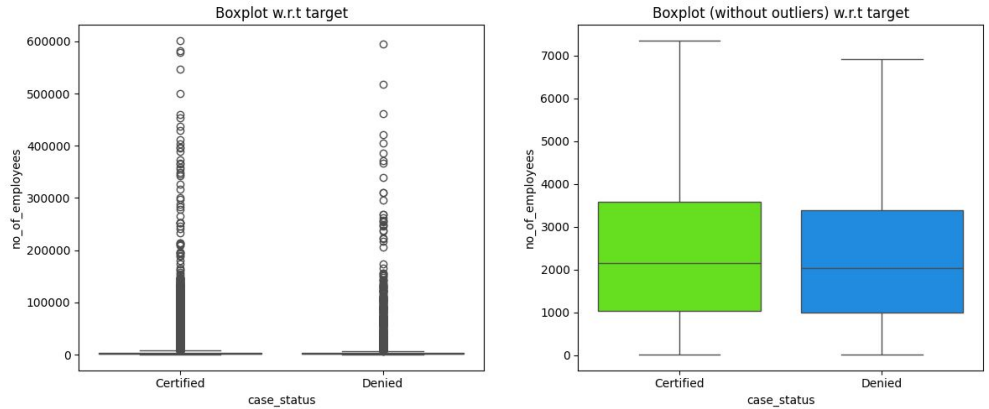


Visa applicants with higher prevailing wage have a higher chance of getting certified.

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EDA Results - Bivariate analysis - Distribution plot wrt target

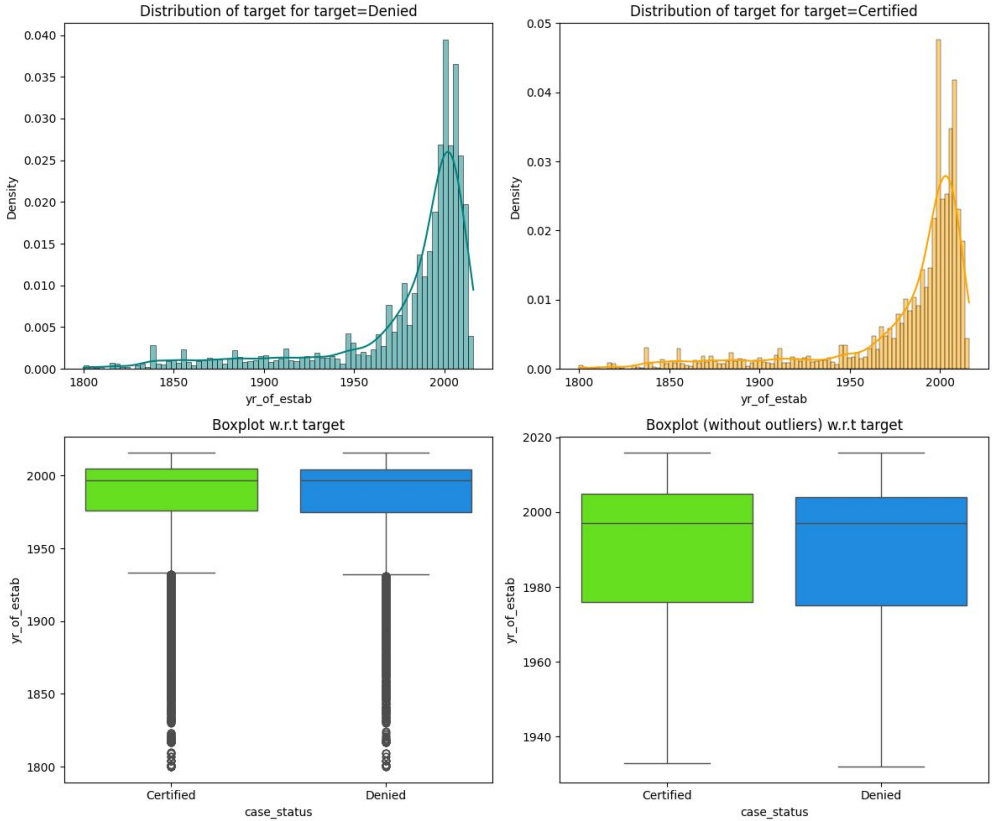




More number of employees indicates more certifications.

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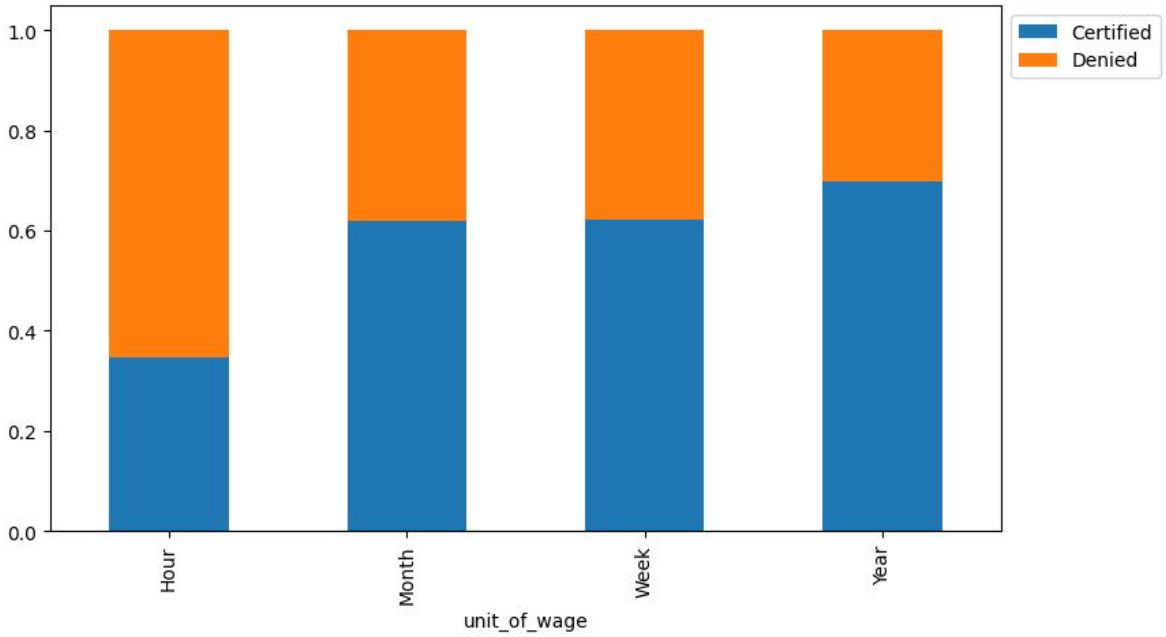
EDA Results - Bivariate analysis - Year of establishment



Certifications may not depend on year of establishment.

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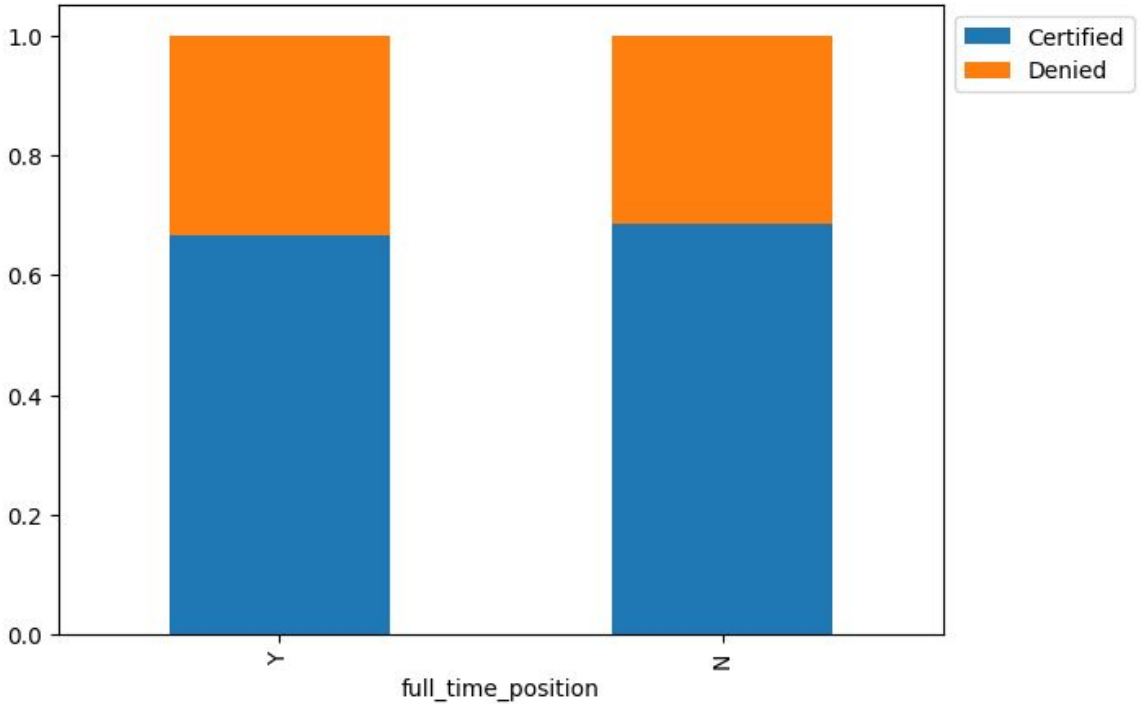
EDA Results - Bivariate analysis - Unit of wage



Employees paid by the hour have lower rates of certifications.

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EDA Results - Bivariate analysis - Full time position



Certifications are not dependent on whether employees have a full time position.

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Model building - Original data

```
Training Performance:

Bagging: 0.9886592370759487
Random Forest: 1.0
AdaBoost: 0.8201575658788373
Gradient Boosting: 0.8284460863541907
XGBoost: 0.8958799403430275

Validation Performance:

Bagging: 0.7722246793454224
Random Forest: 0.803421200224341
AdaBoost: 0.8177474402730376
Gradient Boosting: 0.8249513753820505
XGBoost: 0.8056911703166411
```

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Model building - Oversampled data

```
Training Performance:

Bagging: 0.9873990943099035
Random Forest: 0.9999020568070519
AdaBoost: 0.7992659893478942
Gradient Boosting: 0.8068414203383528
XGBoost: 0.8740713050792019

Validation Performance:

Bagging: 0.7664814536717225
Random Forest: 0.7970702283498492
AdaBoost: 0.8180438681197406
Gradient Boosting: 0.8156058664388438
XGBoost: 0.8084388185654009
```

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Model building - Undersampled data

```
Training Performance:

Bagging: 0.9808497905445841
Random Forest: 0.9999015263417036
AdaBoost: 0.7075129038424776
Gradient Boosting: 0.7275157532938705
XGBoost: 0.8706341463414634

Validation Performance:

Bagging: 0.710274087775349
Random Forest: 0.739038990454619
AdaBoost: 0.7696856520385932
Gradient Boosting: 0.7801992528019925
XGBoost: 0.7442533354766114
```

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Model Performance Summary

- Overview of the final ML model and its parameters
- Summary of most important features used by the ML model for prediction
- Summary of key performance metrics for training and test data of all the models in tabular format for comparison

***Note**: You can use more than one slide if needed*

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Training performance comparison

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned	
Accuracy	0.778519	0.749084	0.751112	0.758242	
Recall	0.917826	0.876200	0.889128	0.977865	
Precision	0.786290	0.776697	0.772530	0.742084	
F1	0.846981	0.823454	0.826738	0.843813	

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Testing performance comparison

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned	
Accuracy	0.746468	0.739403	0.741562	0.714286	
Recall	0.893067	0.873972	0.886604	0.949177	
Precision	0.766129	0.767940	0.764244	0.715773	
F1	0.824742	0.817532	0.820889	0.816115	

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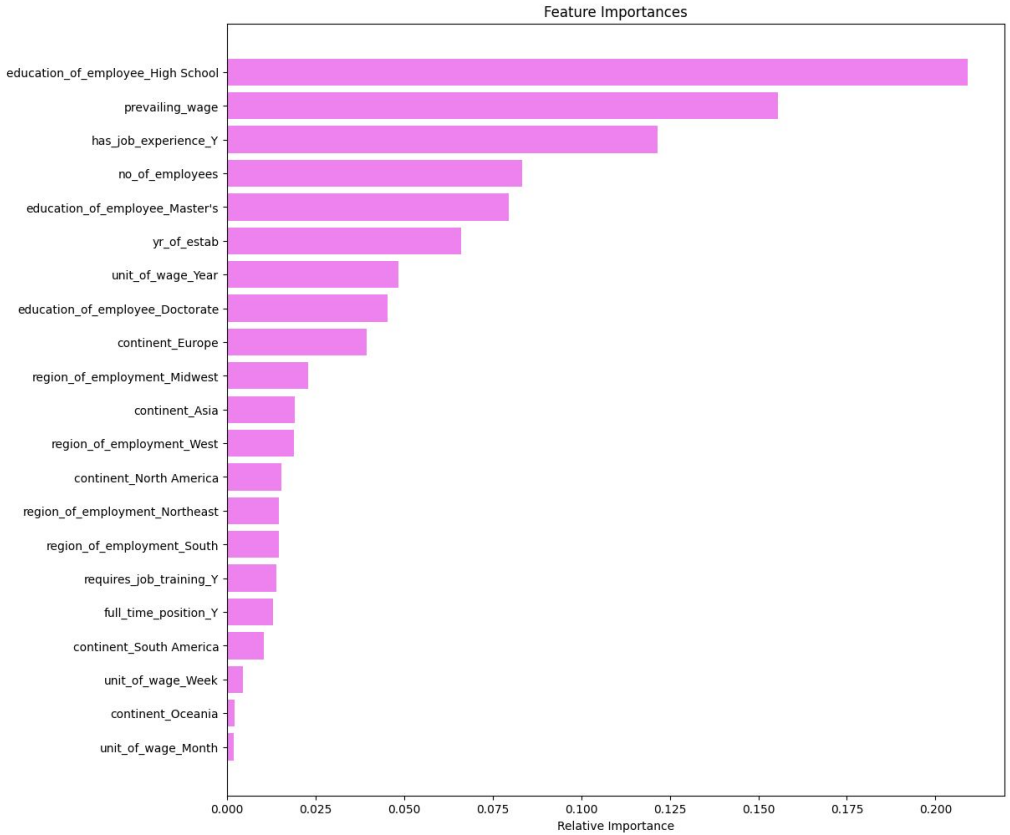
Best performing model

	Accuracy	Recall	Precision	F1
0	0.746468	0.893067	0.766129	0.824742

- Best performing model is : Tuned Random Forest
- Recall score- 0.893067 indicates that the model correctly identifies most true positives.
- Precision score - 0.766129 indicates that positive predictive value is good and results are accurate.
- F1 score - 0.824742 indicates that the precision and recall have a good balance.

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Feature Importances



The top three features that determine certifications in the Tuned Random Forest model (best chosen model) are:

1. Education of employee
2. Prevailing wage
3. Job experience

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